

Requirement Analysis

Project: Smart Parking System

Development Methodology: Agile Extreme Programming (XP)

Frontend: Flutter

1. Project Overview

The **Smart Parking System** is a simple Flutter-based web application that helps users find and reserve available parking spaces. Users can register, log in, view available parking slots, make reservations, cancel bookings, and view their booking history. The project is developed following the Agile Extreme Programming (XP) methodology with Git and GitHub for collaborative development.

2. Problem Statement

Finding an available parking space can be time-consuming and inconvenient. Drivers often waste time searching for empty parking spots. A simple mobile application that displays available parking spaces and allows users to reserve them can make parking faster and more convenient.

3. Stakeholders

- User (Driver)
- Parking Lot Owner
- Development Team

4. Project Scope

In Scope

- User registration and login
- View available parking slots

- Reserve a parking slot
- Cancel a reservation
- View booking history
- User profile management
- Display parking slot availability

5. Functional Requirements

1. Users shall be able to register an account.
2. Users shall be able to log in and log out.
3. Users shall be able to view available parking slots.
4. Users shall be able to reserve an available parking slot.
5. Users shall be able to cancel an existing reservation.
6. Users shall be able to view their current reservation.
7. Users shall be able to view their booking history.
8. Users shall be able to edit their profile information.
9. The system shall update parking slot availability after each reservation or cancellation.
10. The system shall prevent multiple users from reserving the same parking slot.

6. Non-Functional Requirements

1. The application should have a simple and user-friendly interface.
2. The application should respond within 2 seconds under normal conditions.
3. User information should be stored securely.
4. The application should run smoothly on Web platforms.
5. The system should provide reliable reservation information.
6. The application should be easy to maintain and update.
7. The application should support multiple users simultaneously.
8. The project should be developed using Flutter with Git and GitHub for version control.

7. User Stories

User Story 1

As a user, I want to create an account so that I can use the parking system.

User Story 2

As a user, I want to log in so that I can access my parking information.

User Story 3

As a user, I want to view available parking slots so that I can choose an empty space.

User Story 4

As a user, I want to reserve a parking slot so that it is available when I arrive.

User Story 5

As a user, I want to cancel a reservation and view my booking history so that I can manage my bookings easily.

8. Assumptions

- Users have access to web browsers.
- Users have an active internet connection.
- Each parking slot can only be reserved by one user at a time.
- Parking slot information is updated by the system.
- Users provide valid registration information.

9. Constraints

- The application is developed using Flutter.
- The project follows the Agile Extreme Programming (XP) methodology.
- Git and GitHub are used for version control and collaboration.
- The project is limited to basic parking reservation features for the laboratory assignment.
- Online payment, GPS navigation, and hardware integration are not included.
- The project is developed collaboratively by three team members using separate Git branches.